

1 MSE Backward Step

Consider the linear function with one trainable parameter β : $f(x) = \beta + x$.

The table below shows the expected (i.e. target) values for the inputs in the left column:

inputs	targets
0	1
3	4
4	5
5	6

a)

Assume $\beta = 0$ and calculate the MSE!

- $MSE = \frac{1}{n} \sum_{i=1}^n (target_i - prediction_i)^2$
- $MSE = \frac{1}{4} * ((1 - 0)^2 + (4 - 3)^2 + (5 - 4)^2 + (6 - 5)^2) = 1$

b)

Use the partial derivative of MSE w.r.t β given $f(x)$ to calculate the gradient for $\beta = 0$:

- $gradient = \frac{1}{n} \sum_{i=1}^n -2(target_i - (input_i + \beta))$
- $gradient = \frac{1}{4} * ((-2 * (1 - 0)) + (-2 * (3 - 4)) + (-2 * (4 - 5)) + (-2 * (5 - 6)))$
- $gradient = \frac{1}{4} * ((-2) + (-2) + (-2) + (-2))$
- $gradient = \frac{-8}{4}$
- $gradient = -2$

c)

Perform one parameter update of β , using a learning rate of 0.1 and the previously calculated gradient:

- $\beta_{new} = \beta_{old} - (-2 * 0.1)$
- $\beta_{new} = 0 - (-2 * 0.1)$
- $\beta_{new} = 0 - (-0.2)$
- $\beta_{new} = 0.2$

New MSE:

Perform one parameter update of β , using a learning rate of 0.1 and the previously calculated gradient:

- $MSE = \frac{1}{n} \sum_{i=1}^n (target_i - prediction_i)^2$
- $MSE = \frac{1}{4} * ((1 - 0.2)^2 + (4 - 3.2)^2 + (5 - 4.2)^2 + (6 - 5.2)^2)$
- $MSE = \frac{1}{4} * (0.64 + 0.64 + 0.64 + 0.64)$
- $MSE = 0.64$